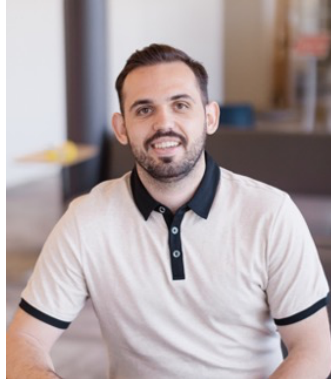




Chris Vellucci

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Curriculum Vitae Content

- A. Formal Education
- B. Academic Distinction and Awards
- C. Academic and Non-Academic Positions Held
- D. Research/Scholarly Activities or Creative Work

A. Formal Education



Ph.D. Applied Health Sciences, Biomechanics (2022-Present)

Brock University, Ontario
Funded by Mitacs Accelerate Entrepreneur
Supervisor: Dr. Shawn Beaudette
Doctoral Dissertation: The development of a AI-driven tool for coaching and rehabilitation.



M.Sc. Applied Health Sciences, Biomechanics (2020-2022)

Brock University, Ontario
Supervisor: Dr. Shawn Beaudette
Master's Thesis: An exploratory study evaluating the effectiveness of a data-driven approach to identify coordinative features that are associated with sprint velocity.



B.Sc. Kinesiology w/ Minor in Ergonomics and Injury Prevention (2016-18)

University of Waterloo, Ontario
Directed Study Supervisor: Dr. Wade Wilson
Directed Study: Exploring athlete buy-in through the coach-athlete relationship.



Diploma in Health Wellness and Fitness with Honours (2014-2016)

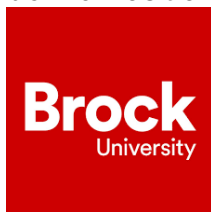
Mohawk College, Ontario

B. Awards/Distinctions

Scholarships and Fellowships		
2025-pres	Queen Elizabeth II Graduate Scholarship in Science and Technology	\$15,000
2025	Mitacs Business Strategy Internship – Lab2Market Launch	\$15,000
2023-pres.	Mitacs Accelerate Entrepreneur	\$180,000
2022-25	Ontario Graduate Scholarship <i>Declined in 25' for QEII</i>	\$45,000
2022-23	Brock VMPI – Research	\$25,000
2022	NSERC Idea to Innovation – Lab2Market Validate	\$20,000
2022-23	ISEK John V. Basmajian Memorial Award	\$750
Honours and Awards		
2021	XSENS University Biomechanics Challenge Finalist	\$0

C. Academic and Non-Academic Positions Held

Academic Positions Held



Graduate Teaching Assistant (Sept 2020-Present)

Department of Kinesiology, Faculty of Applied Health Sciences
Brock University, 1812 Sir Isaac Brock Way, St. Catharines, ON, CAN, L2S 3A1
Courses: KINE 1P98, KINE 1P88, KINE 3P91, KINE 4P91, 4P10

Professional Activities



Founder and President (Sept 2022-Present)

Chalk Monkey Labs, 141 Catherine St, Hamilton, ON, CAN, L8N 2J7
Website: www.cmlabs.ca



Lead Strength and Conditioning Coach (Sept 2021-Aug 2022)

Ridley College, 2 Ridley Rd, St. Catharines, ON L2R 7C3
Website: www.ridleycollege.com



Lead Strength and Conditioning Coach (Aug 2020-Sept 2021)

Brock Sports Performance, 1812 Sir Isaac Brock Way, St. Catharines, ON, CAN, L2S 3A1
Website: www.brocku.ca/bsp/



Sport Science Intern (Jan 2020-March 2020)

University of Minnesota Hockey, 1901 4th St SE, Minneapolis, MN, USA, 55455

Website: www.gophersports.com



Ergonomic Consultant (May 2019-Oct 2019)

Options Inc, 22 Westmount Rd, Guelph, ON, CAN, N1H 5H2

Website: www.oிweb.com



Personal Trainer (Jan 2019-May 2019)

Goodlife Fitness, 100 City Centre Dr, Mississauga, ON, CAN, L5B 2C9

Website: www.goodlifefitness.com

D. Research/Scholarly Activities

1. Publications

Overview

Articles Published in Peer-Reviewed Journals	6
Articles in Prep or Currently Under Peer-Reviewed	3
Peer-Reviewed Conference Proceedings	10
Non-Peer-Reviewed Conference Proceedings	3
Invited Oral Presentations	8

Papers in Peer-Refereed Journals

1. **Vellucci, C.L.**, Beaudette, S. (2024). Multijoint Coordination Contributes to the Minimization of Frontal Plane Center-of-Mass Displacement in Maximal Velocity Sprinting. *Journal of Applied Biomechanics*, 41(6), 561-570. [DOI](#)
2. Ratke, E., McMaster, H., **Vellucci, C.L.**, Larson, D., Holmes, M., Beaudette, S. (2024). Ability of a passive back support exoskeleton to mitigate fatigue related adaptations in a complex repetitive lifting task. *Journal of Biomechanics*, 181, 112553. [DOI](#)
3. Vellucci, C.L., Beaudette, S. (2024). Whole-body coordinative strategies contribute to the minimization of centre of mass displacement in maximal velocity sprinting. *ISBS Proceedings Achieve*, 42(1), 954 [DOI](#)
4. Alano, C., **Vellucci, C.L.**, Beaudette, S. (2024). The effect of neuromuscular fatigue on the spatiotemporal coordination of rowing. *Applied Sciences*, 14(16), 6907 [DOI](#)

Contribution: CLV was involved in the design of protocol, acquisition of data, analysis of the data, and drafting/revision of the manuscript.

5. Wator, J., Battis, A., **Vellucci, C.L.**, Beaudette, S. (2024) The Acute Effects of Kinesio-Taping on Movement Kinematics and Muscle Co- Activation in Rowing

Athletes. *Journal of Back and Musculoskeletal Rehabilitation*, 37 (3), 751-760
[DOI](#)

Contribution: CLV was involved in the design of protocol, acquisition of data, analysis of the data, and drafting/revision of the manuscript.

6. **Vellucci, C.L.**, Beaudette, S.M. (2023). A need for speed: objectively identifying full-body kinematic and neuromuscular features associated with faster sprint velocities. *Frontiers in Sport & Active Living*, 4:1094164. [DOI](#)

Contribution: CLV was involved in the design of protocol, acquisition of data, analysis of the data, and drafting/revision of the manuscript.

Papers Delivered at Professional Meetings:

Peer Reviewed:

1. Ratke, E., McMaster, H., **Vellucci, C.L.**, Larson, D., Holmes, M., Beaudette, S. (2024). Ability of a passive back support exoskeleton to mitigate fatigue related adaptations in a complex repetitive lifting task. (Accepted, Poster)
2. Brooks J.S*., Ratke E.J., Anzovino D.C., Ulona P.A., Guenther A, Patel V, **Vellucci C.L.**, Schroeder R.T., Beaudette S.M., Black A.M. Pinning Down Head Kinematics Across Wrestling Maneuvers. *XXX Congress of the International Society of Biomechanics. Stockholm, Sweden. July 2025* (Accepted, Oral).
3. Brooks J.S*., Anzovino D.C., Ratke E.J., Ulona P.A., Guenther A, Patel V, **Vellucci C.L.**, Schroeder R.T., Beaudette S.M., Black .A.M. A Hard Take Down: Can Markerless Motion Capture Track Wrestling Head Impacts? *XXX Congress of the International Society of Biomechanics. Stockholm, Sweden. July 2025.* (Accepted, Oral)
4. **Vellucci, C.L.***, Beaudette, S.M. (2024). Whole body coordinative strategies minimize frontal place centre of mass displacement during maximal velocity sprinting. *International Society of Biomechanics in Sport, Salzburg, AUS* (Accepted, Oral)
5. **Vellucci, C.L.***, Beaudette, S.M. (2024). Whole body coordinative strategies minimize frontal place centre of mass displacement during maximal velocity sprinting. *Canadian Society of Biomechanics, Edmonton, AB, CAN* (Accepted, Oral)
6. **Vellucci, C.L.***, Beaudette, S.M. (2023). A need for speed: Objectively identifying kinematic strategies associated with faster sprint velocities. *American Society for Biomechanics Congress, Knoxville, TN, USA* (Accepted, Oral).
7. **Vellucci, C.L.***, Beaudette, S.M. (2023). A data-driven approach to identifying key coordinative differences between novice and advanced sprinters. *North American Congress on Biomechanics, Ottawa, ON, CAN* (Accepted, Oral).
8. Alano C.*, **Vellucci, C.L.**, Battis, A., Beaudette, S.M. (2022). The effect of neuromuscular fatigue on the spatiotemporal coordination of rowing. *North American Congress on Biomechanics, Ottawa ON, CAN* (Accepted, Poster).
9. Wator, J*., Battis, A., **Vellucci, C.L.**, Beaudette, S.M. (2022). The acute effects of kinesio- taping on trunk muscle activation in rowing athletes. *North American Congress on Biomechanics, Ottawa ON, CAN* (Accepted, Poster).

10. **Vellucci, C.L.***, Dunne, C.*, Mannella, K.*, Battis, A.* (2021). Using a data driven approach to identify difference in squat and bench press technique with wearable sensors. XSens University Biomechanics Challenge (MSc Work, Oral Presentation, International Competition)

Non-Peer-Reviewed:

1. **Vellucci C.L.***, Beaudette, S.M. (2023). A need for speed: objectively identifying kinematic strategies associated with faster sprint velocities. *Ontario Biomechanics Conference*, Waterloo ON, CAN. (MSc work, Oral).
2. **Vellucci, C.L.***, Beaudette, S.M. (2022). Using a data driven approach to identify key neuromuscular features that are correlated with sprint velocity. MNK Graduate Student Conference (MSc work, Oral).
3. **Vellucci, C.L.***, Beaudette, S. (2021). Motor control of the lumbo-pelvic region during sprinting: A missing link in the sprint performance literature? MNK Graduate Student Conference (MSc work, Oral).

**Corresponds to presenting author*

Invited Oral Presentations

1. **Vellucci, C.L.** (2024). Strength and Power Development Considerations for Athletes. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
2. **Vellucci, C.L.** (2024). Biomechanics of Resistance Training. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
3. **Vellucci, C.L.** (2024). Plyometrics for Improved Sport Performance. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
4. **Vellucci, C.L.** (2024). Periodization for Athletes. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
5. **Vellucci, C.L.** (2024). Fundamentals of Sprint Training. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
6. **Vellucci, C.L.** (2024). Energy System Development for Athletic Populations. Brock University: KINE 4P91 (Guest Lecture Series, Brock University)
7. **Vellucci, C.L.** (2022). Coaching Sprint Technique. Brock University: KINE 3P91 (Guest Lecture, Brock University)
8. **Vellucci, C.L.** (2022). Leveraging machine learning to enhance coaching practices. *Research Café: Sports Edition*, St. Catharines ON, CAN.

Works Submitted or in Prep (Final Stages for Publication)

1. **Vellucci, C.L.**, Beaudette, S. (2024). Can 3D kinematics be derived from a single 2D video?
Status: Preparing for submission to Computer Methods in Biomechanics and Biomedical Engineering
Contribution: CLV was involved in the design of protocol, acquisition of data, analysis of the data, acquisition of funding, and drafting/revision of the manuscript.

2. **Vellucci, C.L.**, Doctor, I., Ratke, E.J., Holmes, M., Beaudette, S. (2026). The use of transfer learning to take estimations of 3D kinematics via 2D video from walking to running.

Status: Final Stages of Data Acquisition

Contribution: CLV was involved in developing the scripts for the markerless motion capture algorithm, and statistical analysis

3. **Vellucci, C.L.**, Doctor, I., Holmes, M., Beaudette, S. (2026). Concurrent evaluation of marker, markerless, and wearable motion capture systems to detecting changes in running gait caused by speed, exertion, and pain.

Status: Final Stages of Data Acquisition

Contribution: CLV was involved in developing the scripts for the markerless motion capture algorithm, and statistical analysis